



AI Programming: Assignment4Part2

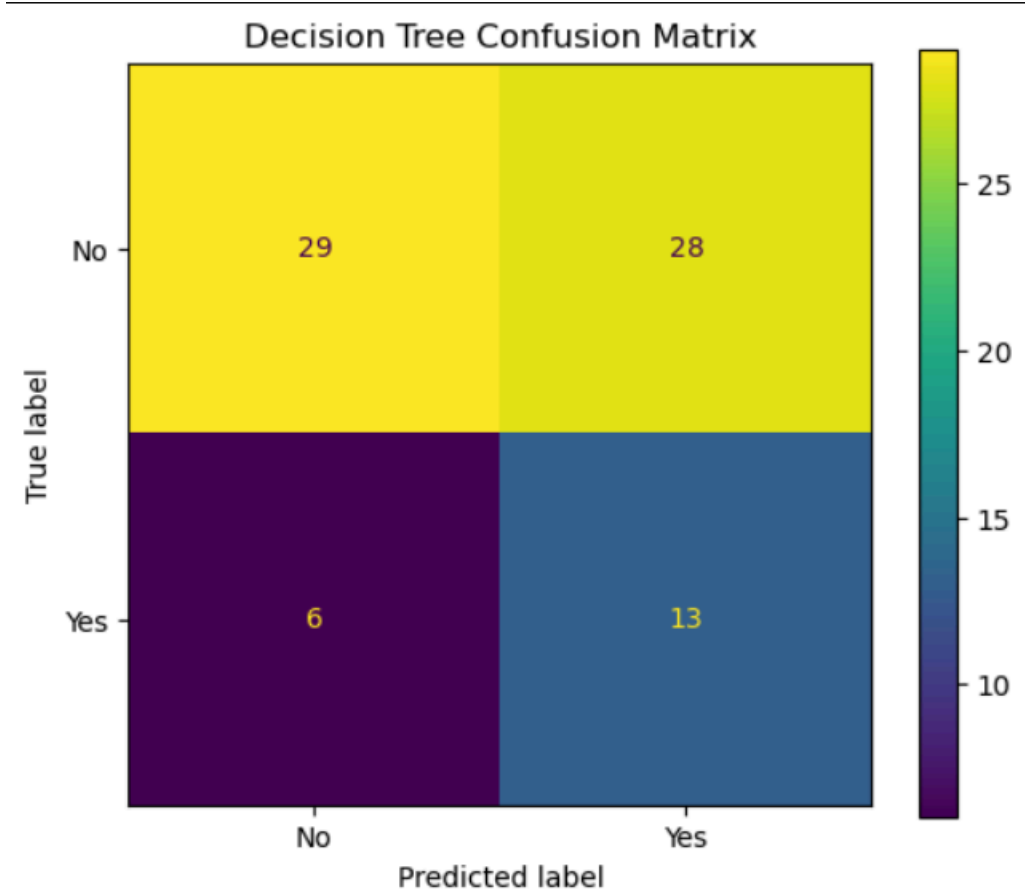
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Results and Discussion

A Decision Tree classification model was developed to predict **Booking_Cancelled**, with the objective of identifying hotel reservations that are likely to be cancelled before the customer's arrival date. Accurate cancellation predictions can help hotels improve occupancy planning, optimize room allocation, and reduce revenue losses associated with last-minute cancellations.

Confusion Matrix Analysis



The confusion matrix provides a detailed breakdown of the model's predictions on the testing dataset. Out of 76 bookings evaluated, the model correctly classified 42 bookings and incorrectly classified 34 bookings.

The model correctly identified 29 bookings that were not cancelled (True Negatives) and 13 bookings that were cancelled (True Positives). However, it incorrectly predicted 28 bookings as cancelled when they were not actually cancelled (False Positives), and it failed to identify 6 bookings that were eventually cancelled (False Negatives).

The relatively large number of false positives indicates that the model tends to predict cancellations more frequently than necessary. While this may lead to unnecessary

precautionary actions by hotel management, it is generally less costly than missing actual cancellations. The relatively low number of false negatives suggests that the model is reasonably effective at identifying bookings that are likely to be cancelled.

Classification Performance

The Decision Tree model achieved an overall accuracy of **55.26%** on the testing dataset. This indicates that slightly more than half of all bookings were classified correctly.

The model achieved a **precision score of 31.71%**, meaning that when the model predicts a booking will be cancelled, it is correct approximately one-third of the time. This relatively low precision reflects the large number of false positive predictions.

The **recall score of 68.42%** demonstrates that the model successfully identified more than two-thirds of the actual cancelled bookings. From a business perspective, recall is particularly important because failing to identify likely cancellations may result in empty rooms and lost revenue opportunities.

The model achieved an **F1-score of 43.33%**, indicating that there is room for improvement in balancing precision and recall. The relatively low F1-score is primarily driven by the model's tendency to generate false positive predictions.

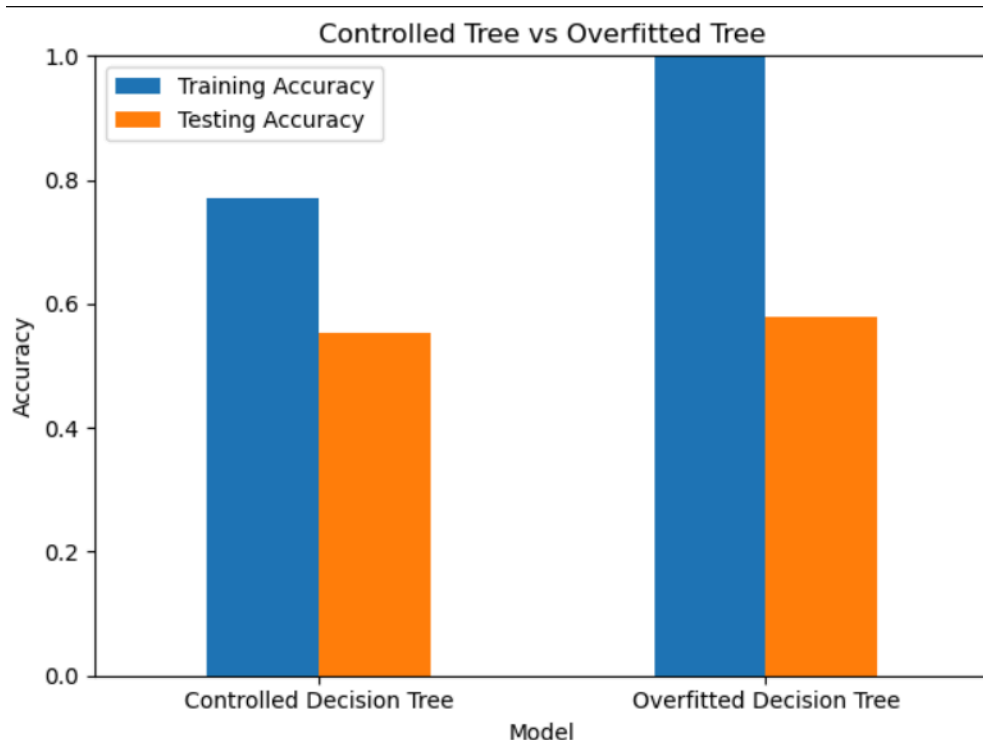
Class-Level Performance

The classification report provides additional insight into the model's performance for each booking outcome.

For bookings that were not cancelled (Class 0), the model achieved a precision of 0.83, recall of 0.51, and F1-score of 0.63. This indicates that when the model predicts a booking will not be cancelled, it is often correct, although it misses a significant number of actual non-cancelled bookings.

For cancelled bookings (Class 1), the model achieved a precision of 0.32, recall of 0.68, and F1-score of 0.43. The higher recall demonstrates that the model is reasonably successful at identifying cancellations, but the low precision indicates that many bookings predicted as cancelled are ultimately not cancelled.

Overfitting Analysis



To evaluate model generalization, a controlled Decision Tree was compared with an unrestricted Decision Tree.

The controlled Decision Tree achieved a training accuracy of **76.97%** and a testing accuracy of **55.26%**. The decrease in performance from training to testing data suggests that the model does not generalize particularly well to unseen bookings.

The unrestricted Decision Tree achieved **100% training accuracy**, demonstrating perfect performance on the training data. However, its testing accuracy was only **57.89%**, indicating severe overfitting. The model memorized patterns in the training dataset rather than learning relationships that generalize effectively to new data.

Although the overfitted model achieved slightly higher testing accuracy, its perfect training performance strongly suggests that it is not a reliable model for real-world deployment.

Business Implications

The results indicate that the Decision Tree model has some ability to identify bookings that are likely to be cancelled, particularly due to its relatively strong recall score. This capability can help hotel managers anticipate cancellations and make proactive decisions regarding room inventory management and overbooking strategies.

However, the model also produces a substantial number of false positive predictions. As a result, hotel managers should use the model as a decision-support tool rather than relying on it

exclusively. Additional model tuning, feature engineering, or more advanced machine learning techniques may be required to improve prediction accuracy.

Conclusion

The Decision Tree model demonstrated moderate success in predicting hotel booking cancellations. While the model was able to identify a majority of cancelled bookings, its overall accuracy and precision were relatively low. The comparison between the controlled and overfitted Decision Trees highlights the importance of limiting model complexity to improve generalization.

Overall, the model provides useful insights into booking cancellation behavior but would benefit from further optimization before being used as a primary decision-making tool in a hotel management environment.